United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Trimester: Summer 2024 Marks: 50 Time: 2 Hours

Code: CSE 3411 Course Title: System Analysis & Design

“Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.”

Answer all the following questions:

Question 1: [CO3] [16]

Consider the following scenario:

The university is deploying a new "SecureLocker" system across the campus, offering smart lockers that provide students, faculty, and staff a secure and flexible storage solution. Users can interact with these lockers through a mobile app, which allows them to reserve lockers remotely, specify the duration of use, and unlock the lockers via their smartphones. In addition to reserving and accessing lockers, users can share locker access with trusted individuals, such as friends or colleagues. The system operates with three distinct types of users. Students, the primary users of the lockers, can reserve lockers, manage their reservation times, and share access with others. They also receive notifications regarding their lockers, such as reminders when their reservation is about to expire or alerts in case of any unauthorized access attempts. Maintenance technicians, responsible for ensuring the smooth operation of the system, handle tasks like diagnosing issues, performing repairs, and conducting regular system checks.

Administrators, who oversee the entire SecureLocker network, have control over all lockers, manage the inventory, access reports on locker usage, and have the authority to impose temporary system-wide locks for maintenance or emergencies. The system is integrated with the university’s identity management platform to ensure only authorized users can access the lockers. Furthermore, it tracks all locker-related activities, allowing administrators to review logs for security or maintenance purposes. Considering the above scenario, answer the following questions:

a) Design a Use Case Diagram for the "SecureLocker" application. Ensure that you include at least one include or extend relationship. [4]

b) Draw the Class Diagram and CRC card for the "SecureLocker" application by identifying the main classes and their relationships. [4]

c) Create a State Diagram that depicts the lifecycle of a locker in the "SecureLocker" system. [4]

d) Draw a Sequence Diagram that demonstrates the process of a student reserving a locker, sharing access with another user, and unlocking it for retrieval. [4]

Question 2: [CO4] [8]

a) Imagine you and your team are considering a new IoT based project in the field of renewable energy. The initial investment required for the project is $20,000, which will cover the costs of initial installation. You spent $10000 for equipment and installation in the 2^{nd} year. In the 5th year, you had to bear the expenses for the maintenance is around $10000. During the five years project the average HR cost becomes $10000 per year. The projected revenues are as follows: Year 1: $20,000, Year 2: $30,000, Year 3: $50,000, Year 4: $60,000, and Year 5: $70,000. Consider the rate of interest is 12%.

Determine whether the project is financially sustainable or not (profit/loss) for your team using the **Net Present Value** following method: [4]

b) Mention the different types of feasibility analysis. What is the concept of technical feasibility and why it is important to analyze before an IT product development. [2+2]

Question 3: [CO3] [16]

a) Write down some guidelines of UI design in developing a software product. Draw the proposed UI Design of "Homepage/Landing-page" for the project described in question 1 also mention guideline/s you have followed in the proposed UI design. [2+3+2]

b) "An effective SRS maintains fundamental 3C properties: Completeness, Correctness, and Consistency" – describe briefly. [3]

c) Determine the functional and non-functional features/requirements for the scenario mentioned in Question 1. Write down any two sample non-functional requirement's specification in details. [3+3]

Question 4: [CO2, CO3, CO4] [10]

Write down an SRS document and present it as a group having the following parts on your software project work covering: Introduction, Problem Statement, Information Gathering and System Study, System Analysis, Feature list fixation with Functional and Non-functional requirements, Different Feasibility Analysis, System design (UML diagrams), UI design through Figma software, implementation plan etc. (**no need to answer this question in the examination**, it has already been evaluated in classroom activities). [10]